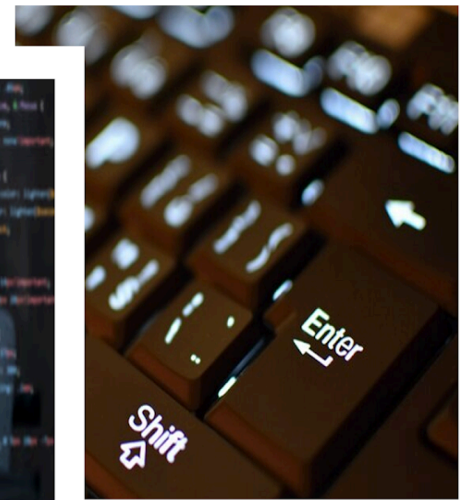


Computer Awareness

STUDY NOTES

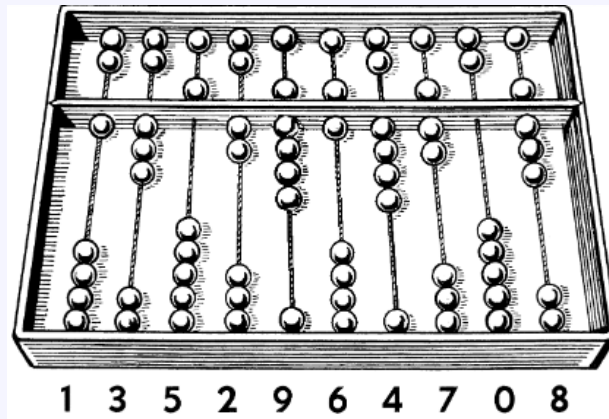
History Of Computer



History Of Computers

1. Ancient Counting Tools and Early Calculators

- **Abacus (750 BC):** The abacus, meaning "board" or "calculating table," dates back to around 750 BC. The Egyptians designed an abacus with a rectangular frame, rods, and a divider, featuring beads of different colors for counting. Similar devices were made by Hindus in India on sand and by Greeks on stones or slabs. The **Chinese later perfected the abacus.**
- It typically had 10 columns. Addition or subtraction was performed by moving the beads sideways. The place value of digits and the position of beads facilitated the calculation process.
- Egyptians, Hindus, Greeks, Chinese.



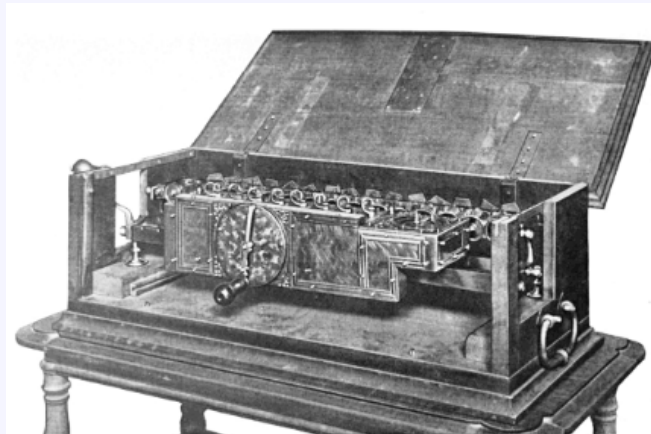
- **Napier's Bones (1550-1617):** Created by John Napier between 1550 and 1617, are a set of rods used for multiplication and division through a method of lattice multiplication.



- **Pascal's Adding Machine (1642):** Invented by Blaise Pascal in 1642, was an early mechanical calculator capable of performing addition and subtraction.



- **Leibnitz Calculator (1673):** Developed by Gottfried Wilhelm Leibniz in 1673, was an advanced mechanical calculator that could perform addition, subtraction, multiplication, and division.



2. Innovative Inventions with Punched Cards and Early Mechanical Computers

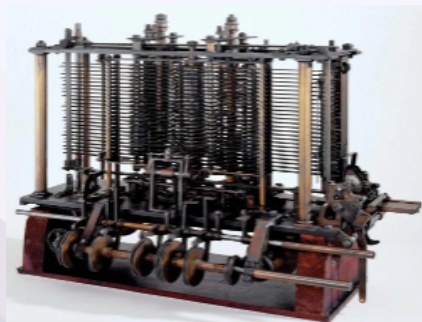
- **Jacquard's Loom (1801):** Invented by Joseph Marie Jacquard in 1801, was a mechanical loom that used punch cards to control the weaving of complex patterns, influencing the development of programmable machines.



- **Babbage's Difference Engine (1822):** Designed by **Charles Babbage (Father of Computer)** in 1822, was an early mechanical computer intended to automatically compute and print mathematical tables. It is the foundation for modern computers.



- **Babbage's Analytical Engine (1833):** Conceptualized by Charles Babbage in 1833, was a proposed mechanical general-purpose computer. It was designed to perform any calculation or algorithm through the use of punched cards, and it contained components analogous to the modern CPU, memory, and I/O devices, making it a precursor to modern computers.

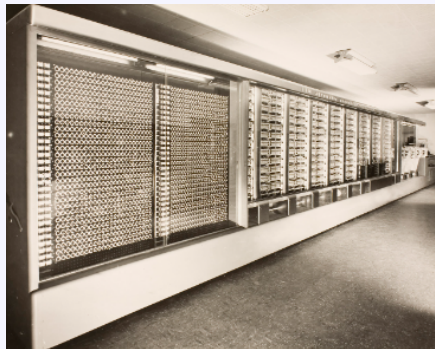


3. Electromechanical and Electronic Breakthroughs

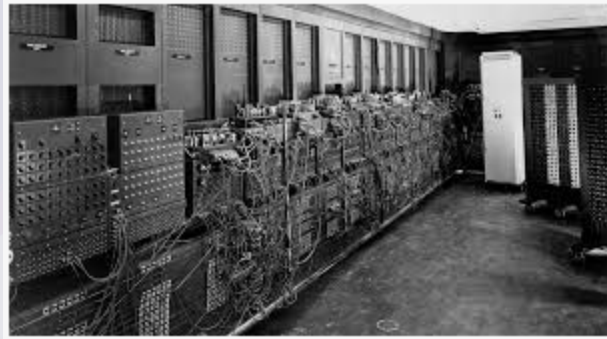
- **Hollerith's Card Reader (1877):** An early data processing machine by Herman Hollerith, used punched cards for the 1890 U.S. Census to quickly tabulate data.



- **Mark-I (1937-1943):** An electromechanical computer designed by Howard Aiken and built by IBM, notable for automating long computations and aiding in World War II calculations.



- **ENIAC (1946):** The **Electronic Numerical Integrator And Calculator** is the first **general-purpose electronic digital computer** by Eckert and Mauchly, significantly faster than previous machines, and revolutionized computing. First electronic computer using vacuum tubes; used for military calculations.



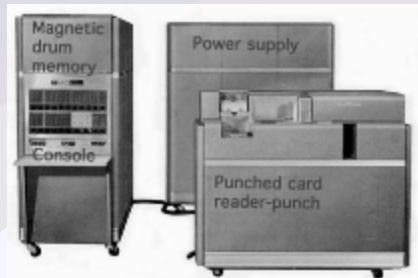
- **EDVAC (1949 AD):** The **Electronic Discrete Variable Automatic Computer**, designed by **John von Neumann**, was one of the first computers to use stored-program architecture, which allowed for more versatile and efficient computation compared to its predecessors.



- **UNIVAC (1951 AD):** The **Universal Automatic Computer**, developed by **J. Presper Eckert and John Mauchly**, was the **first commercially produced computer in the United States**. It was used widely for business and government applications, marking the beginning of the computer age for commercial data processing.



- **IBM 650 (1955 AD):** The IBM 650 was the first mass-produced computer, popular for its relatively affordable cost and widespread use in education and business.

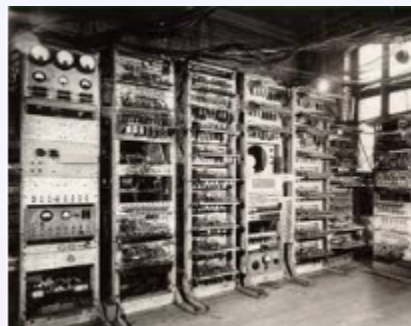


- **IBM 700 Series (1955 AD):** The IBM 700 series were large-scale computers designed for **scientific and industrial calculations**, becoming pioneers in the era of mainframe computing.



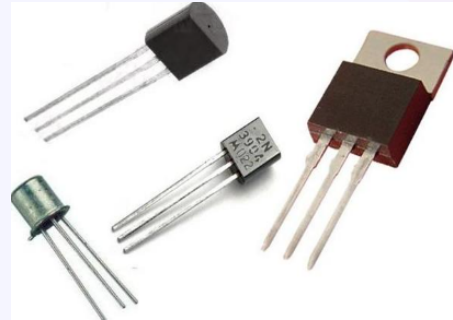
4. Generations of Computers

- **First Generation (up to 1954):** Used **vacuum tubes**; large, slow, costly, and consumed a lot of power.
- **Examples:** ENIAC, EDVAC, IBM 650, IBM 702, etc

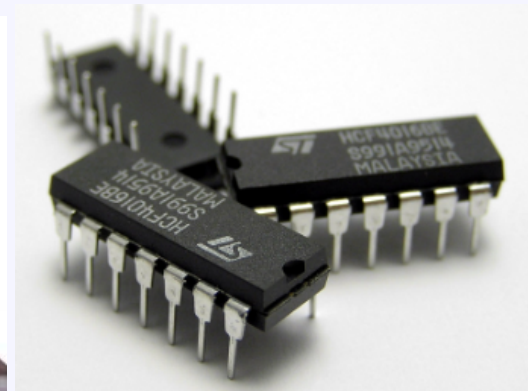


- **Second Generation (1955-1964):** **Transistors** replaced vacuum tubes, enabling smaller, faster computers with lower power consumption.

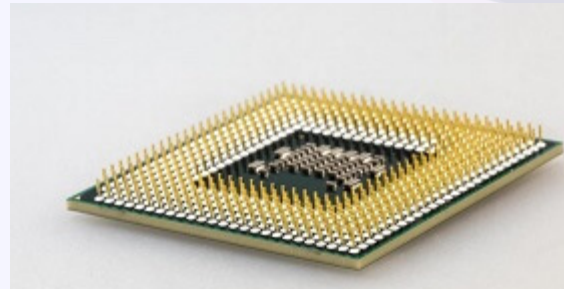
- **Examples:** IBM 1620, IBM 7094, CDC 1604, CDC 3600, and UNIVAC 1108



- **Third Generation (1965-1974): Integrated Circuits (ICs)** led to more compact, efficient systems; introduced database management, mainframes.
- **Examples:** IBM System/360, DEC PDP-8, CDC 6600, Honeywell-6000 series, and IBM-370/168



- **Fourth Generation (1975-2002): VLSI (Very large-scale integration)** chips with thousands of transistors, led to personal computers, laptops, and multimedia.
- **Examples:** IBM 1401, STAR 1000, CRAY-X-MP, PDP 11 etc.



- **Fifth Generation (Present & Future): Ultra-Large Scale Integration (ULSI)** with **AI capabilities**, natural language processing, and self-learning.
- Examples: Notebooks from IBM, PCs of pentium, PCs of pentium, Workstations of SUN, IBM SP/2, Supercomputers with PARAM



5. Types of Computers

- **Mechanical Computers:** Early non-electric devices like Abacus, Pascal's machine.
- **Electronic Computers:** ENIAC as a pioneer, progressing to modern digital electronics.
- **Analog Computers:** Analog computers operate using continuously varying electrical voltages for data and calculations. Devices like voltmeters and ammeters measure these values. They perform parallel calculations, making them faster but prone to errors. These computers are commonly found in scientific and engineering equipment in educational labs.

- **Digital Computers:** Digital computers convert data into binary code and process it. Their data is discontinuous, but they operate at very high speeds, which continue to improve. They are more accurate than analog computers and are utilized in virtually every field of life.
- **Hybrid Computers:** Combine analog and digital, e.g., medical machines.

6. Categories of Modern Computers

- **Mainframe Computers:** Mainframe computers handle large transaction volumes for big databases, ideal for banks, corporations, and government offices. Once room-sized, modern mainframes now look like typical workstations but are still costly. They support many users at once, ensuring data integrity and security. Key uses include payroll, banking, reservations, and government administration, e.g., IBM 4300 series.
- **Minicomputers:** Minicomputers, like IBM AS/400 and VAX 8842, serve as mid-range computers typically used as network servers for tasks such as file sharing, graphics-intensive work, and simulations. They offer a balance between performance, cost, and size, priced lower than mainframes but still costly for general consumers.
- **Supercomputers:** Most powerful, used in research and large-scale calculations, e.g., CRAY series, PARAM, Deep Blue etc.
- **Workstations:** High-performance PCs designed for specific applications like graphics or simulations.
- **Personal Computers (PCs):** Common in homes/offices, low cost, diverse configurations.
- **Laptops:** Portable PCs with low power needs, useful for travel.
- **Handheld Devices:** Small, portable devices, e.g., PDAs, mobile phones.
- **Wearable Computers:** Integrated into clothing, e.g., smartwatches.